

CSCE 883 — Homework 1

K-Nearest Neighbors and Naive Bayes

Learning goals. Implement and evaluate KNN, explain the effect of feature scaling, calculate a Naive Bayes prediction by hand, evaluate a scikit-learn classifier, and audit an imperfect ML pipeline.

General requirements

- Submit one compressed folder containing your code, report (PDF preferred), and any required AI-use disclosure. Your code must run from the submitted folder with relative paths.
- Unless a question explicitly permits a library implementation, write the requested algorithm yourself. You may use standard libraries for data loading, numerical arrays, plotting, and metrics.
- AI coding assistants are permitted. You remain responsible for every line submitted. For Question 5, disclose the tool(s) and prompts used; for all questions, cite any external sources beyond the supplied files.
- Report numerical results to three decimal places. Explain results in your own words; code output alone does not earn full credit.

Question 1 — KNN and feature scaling

Files: Q1_train.csv and Q1_test.csv. Each file contains four numeric features followed by a binary class label in the final column.

- Implement a K-nearest-neighbors binary classifier from scratch using Euclidean distance. Do not use `sklearn.neighbors.KNeighborsClassifier`.
- Train on Q1_train.csv and evaluate once on Q1_test.csv for $K = 1, 2, \dots, 9$. Record the number of correct predictions and accuracy.
- When a vote is tied, predict the class of the single nearest tied neighbor. State this policy in your report.

K	1	2	3	4	5	6	7	8	9
Correct predictions									
Accuracy									

- Repeat the experiment after min–max normalization and after standardization. Fit each transformation using only the training features, then transform the test features with those same training-set parameters. Do not use any test-set statistics while fitting the transformation.

K	1	2	3	4	5	6	7	8	9
Correct predictions									
Accuracy: normalization									
Accuracy: standardization									

- In 150–200 words, identify the best configuration and explain why feature scale can change Euclidean-distance KNN performance. Comment on whether the result is stable across nearby K values.

Definitions. For feature x , min–max normalization is $(x - \text{min_train})/(\text{max_train} - \text{min_train})$. Standardization is $(x - \text{mean_train})/\text{standard_deviation_train}$. Handle a zero-variance training feature explicitly.

Question 2 — Selected research papers on KNN

Select any two of the following papers. These are selected historical papers, not necessarily the latest work on KNN.

- Papernot, N., & McDaniel, P. (2018). Deep k-nearest neighbors: Towards confident, interpretable and robust deep learning. arXiv:1803.04765.
- Jiang, W. (2020). Time series classification: nearest neighbor versus deep learning models. SN Applied Sciences, 2, 1–17.
- Bergman, L., Cohen, N., & Hoshen, Y. (2020). Deep nearest neighbor anomaly detection. arXiv:2002.10445.
- Yang, X., Nan, X., & Song, B. (2020). D2N4: A discriminative deep nearest neighbor neural network for few-shot space target recognition. IEEE Transactions on Geoscience and Remote Sensing, 58(5), 3667–3676.

For each selected paper, write a 250–350 word summary in your own words. State the problem, the main KNN-related idea, the evidence used to evaluate it, and one limitation or open question.

- Include a full citation and a working DOI, arXiv, or publisher link for each paper. You may include one original figure or a clearly attributed/adapted figure with a caption.
- Propose one KNN improvement of your own (150–200 words). Specify the problem it addresses, the mechanism, and an experiment that could falsify or support the idea.

Question 3 — Manual Naive Bayes calculation

Use the car-theft dataset below to classify a Red, Domestic, SUV. Treat Color, Type, and Origin as conditionally independent given the class label (the Naive Bayes assumption).

Example	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

- Show $P(\text{Yes})$ and $P(\text{No})$.
- Show each conditional likelihood needed for $P(\text{Red, Domestic, SUV} \mid \text{Yes})$ and $P(\text{Red, Domestic, SUV} \mid \text{No})$.
- Calculate and compare the two unnormalized posterior scores. Then state the predicted class.
- Use no smoothing for the main calculation. In one or two sentences, explain what Laplace smoothing would do if a required conditional count were zero.

Question 4 — Scikit-learn Naive Bayes on German credit data

Use Q4-train.csv for training and Q4-test.csv for testing. These are comma-separated CSV files with no header row. Each row contains 24 encoded input attributes followed by the class label in the final column. The source files german_credit_data.csv, german.data-numeric.csv, and german.doc are background/reference materials only.

- Load the files correctly, separate the final column as y , and verify the number of rows and columns after loading.
- Choose one scikit-learn Naive Bayes implementation. State which implementation you selected and justify whether its distributional assumptions are a reasonable approximation for these encoded attributes.
- Fit only on Q4-train.csv. Report test-set accuracy, macro-precision, macro-recall, macro-F1, and a labeled confusion matrix for Q4-test.csv.
- Inspect at least five misclassified test examples. Briefly discuss a possible reason for each error based on the encoded-feature representation; do not claim that an encoded number necessarily has continuous numeric meaning.
- Submit reproducible code and its text output. Screenshots alone are not acceptable evidence.

Question 5 — Debugging and validating a classification pipeline

The file `q5_starter_classifier.py` is supplied with this homework. It runs on the Question 1 data, but its results and/or methodology are not fully reliable. **You may use AI tools to investigate and repair it;** you must independently validate every conclusion.

- Identify and correct at least three substantive problems in the starter pipeline. A substantive problem affects correctness, evaluation validity, or the interpretation of results; style-only changes do not count.
- For each problem, report the affected code location, why it is incorrect or methodologically invalid, your correction, and evidence that demonstrates the impact or validity of the correction.
- After fixing the pipeline, compare a corrected KNN model and a justified Naive Bayes model on the same held-out test data. Report accuracy, macro-F1, and labeled confusion matrices.
- Write a 200–500 word reflection: what the starter code did correctly, what was wrong or underspecified, what evidence justified each repair, and which model you would recommend for this dataset and why.
- Append an AI-use disclosure listing the tool(s), prompts or prompt summaries, and how you checked AI-provided claims. Unsupported AI output receives no credit.

Examples of valid investigations. Potential issues may concern data leakage, feature/label separation, distance computation, voting ties, model assumptions, metric calculations, or interpretation. This list is illustrative, not a checklist of required answers. The goal is to learn the traps in applying KNN or Naïve Bayes Classifier to real problems.